Physics 1 Constants

Constants

Wien's displacement constant	$a = 2.90 \times 10^{-3}\,\mathrm{K\,m}$
Boltzmann's constant	$k = 1.38 \times 10^{-23}\,\mathrm{J/K}$
Stefan–Boltzmann constant	$\sigma = 5.67 \times 10^{-8}\,\mathrm{W/m^2K^4}$
Planck's constant	$h = 6.63 \times 10^{-34}\,\mathrm{J\,s}$
Bohr's constant	$B = 2.18 \times 10^{-18}\,\mathrm{J}$
Speed of light in vacuum	$c = 3.00 \times 10^{8}\,\mathrm{m/s}$
Elementary charge	$e = 1.60 \times 10^{-19}\,\mathrm{C}$
Acceleration of gravity	$g = 9.81\,\mathrm{m/s^2}$
Atomic mass unit	$u = 1.66 \times 10^{-27}\,\mathrm{kg}$
Light year	$1\,\mathrm{ly} = 9.46 \times 10^{15}\,\mathrm{m}$

Masses

Electron	5.4858×10^{-4} u $= 9.1094 \times 10^{-31}$ kg
Proton	$1.007\,276$ u $= 1.6726 \times 10^{-27}$ kg
Neutron	$1.008\,665$ u $= 1.6749 \times 10^{-27}$ kg
Hydrogen	$1.007\,825$ u
Deuterium	$2.014\,102$ u
Tritium	$3.016\,049$ u
Helium-3	$3.016\,029$ u
Helium-4	$4.002\,603$ u
Alpha particle	$4.001\,506$ u

Earth

Mass	$5.972 \times 10^{24}\,\mathrm{kg}$
Mean radius	$6371\,\mathrm{km}$
Mean distance from the Sun	$1.496 \times 10^{11}\,\mathrm{m}$
Solar irradiance outside the atmosphere	$1361\,\mathrm{W/m^2}$

Sun

Mass	$1.99 \times 10^{30}\,\mathrm{kg}$
Radius	$6.96 \times 10^{8}\,\mathrm{m}$
Surface temperature	$5780\,\mathrm{K}$
Radiated power	$3.83 \times 10^{26}\,\mathrm{W}$

SI prefixes

10^{-18}	atto	a	10^{-3}	milli	m	10^{3}	kilo	k
10^{-15}	femto	f	10^{-2}	centi	c	10^{6}	mega	M
10^{-12}	pico	p	10^{-1}	deci	d	10^{9}	giga	G
10^{-9}	nano	n	10^{1}	deca	da	10^{12}	tera	T
10^{-6}	micro	µ	10^{2}	hecto	h	10^{15}	peta	P

Physics 1 Formulas

Motion and forces

$v = at + v_0$ $\qquad s = \dfrac{v + v_0}{2}\,t$

$v^2 - v_0^2 = 2as$ $\qquad s = \frac{1}{2}at^2 + v_0 t$

$\Sigma F = 0 \iff v$ constant and rectilinear

$\Sigma\vec{F} = m\vec{a}$ $\qquad F^* = F$ $\qquad G = mg$

$R = \mu N$ $\qquad B = \rho_f V g$ $\qquad L = kv^2$

$\vec{p} = m\vec{v}$ $\qquad \Sigma\vec{F}\cdot t = \Delta\vec{p}$

$m_A v_A + m_B v_B = m_A v_{A0} + m_B v_{B0}$

Mechanical energy and power

$W = Fs\cos\theta$ $\qquad W_{\Sigma F} = \Delta E_k$ $\qquad P = W/t$

$E_k = \frac{1}{2}mv^2$ $\qquad E_p = mgh$ $\qquad E = E_k + E_p$

$\frac{1}{2}mv^2 + mgh = \frac{1}{2}mv_0^2 + mgh_0$

$\eta = \dfrac{\text{useful energy}}{\text{supplied energy}} = \dfrac{\text{useful power}}{\text{supplied power}}$

Pressure and temperature

$p = F/A$ $\qquad T = t + 273$ $\qquad E_k = \frac{3}{2}kT$

$c = \dfrac{Q}{m\,\Delta T}$ $\qquad C = \dfrac{Q}{\Delta T}$ $\qquad \Delta U = W + Q$

Pressure and temperature (cont.)

$COP = \dfrac{Q_{\text{out}}}{E_{\text{electric}}}$ $\qquad COP_{\text{Carnot}} = \dfrac{T_{\text{out}}}{T_{\text{out}} - T_{\text{in}}}$

Waves and radiation

$f = 1/T$ $\qquad v = \lambda f$ $\qquad U = P/A$

$\lambda_{\text{peak}}\cdot T = a$ $\qquad U = \sigma T^4$ $\qquad I = \dfrac{P}{4\pi r^2}$

Atomic and nuclear physics

$E = hf = \dfrac{hc}{\lambda}$ $\qquad E_n = -\dfrac{B}{n^2}$

$hf = E_n - E_m, \quad n > m$ $\qquad hf = B\left(\dfrac{1}{m^2} - \dfrac{1}{n^2}\right)$

$A = Z + N$ $\qquad E_0 = mc^2$ $\qquad \Delta E_0 = \Delta m\cdot c^2$

Electricity

$I = \dfrac{Q}{t}$ $\qquad U = \dfrac{W}{Q}$ $\qquad R = \dfrac{U}{I}$

$W = UIt$ $\qquad P = UI = RI^2 = \dfrac{U^2}{R}$

Series: $\quad R_{\text{tot}} = R_1 + R_2 + \cdots + R_n$

Parallel: $\quad R_{\text{tot}} = \left(\dfrac{1}{R_1} + \dfrac{1}{R_2} + \cdots + \dfrac{1}{R_n}\right)^{-1}$